Wazuh – Threat Detection

WEB – BASED DETECTION

RAMYA RADJESH

Executive Summary

This project demonstrates the detection and monitoring of web application attacks using Wazuh SIEM in a controlled virtualized lab environment.

Multiple web-based attacks including brute force authentication attacks, SQL injection, command injection, cross-site scripting (XSS), and file upload exploitation were executed against the DVWA application. The Apache logs generated by these attacks were collected by the Wazuh Agent and forwarded to the Wazuh Manager for analysis.

The Wazuh SIEM successfully detected malicious activities and generated alerts with appropriate severity levels. However, while detection was effective, the application itself lacked preventive controls such as rate limiting, input validation, and file upload restrictions.

This project highlights the importance of centralized log monitoring, real-time detection, and proactive security controls in protecting web applications from exploitation.

wazuh.

Contents

Wazuh. 1

Wazuh - Agent 3

Wazuh – Manager 5

Wazuh – Dashboard 7

Opensearch 9

wazuh.

Wazuh.

Wazuh is an open-source security monitoring and threat detection platform used for intrusion detection, log analysis, and compliance management. It provides organizations with real-time visibility into security events across endpoints, cloud workloads, and network infrastructure, helping detect vulnerabilities and respond to incidents efficiently.

Key facts

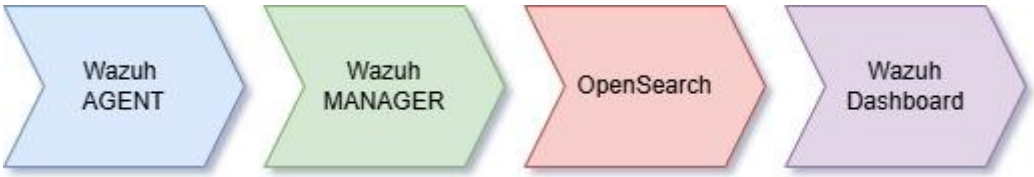
- Initial release: 2015
- Type: Open-source security monitoring platform
- Core components: Wazuh agent, manager, indexer, and dashboard
- Primary use cases: SIEM, XDR, and compliance monitoring
- License: GNU General Public License (GPL v2)

Architecture and Components

Wazuh's modular architecture includes several main components:

- **Agent:** Installed on monitored endpoints to collect system and security event data.
- **Manager:** Processes, analyzes, and correlates collected data for threat detection.
- **Indexer:** Stores event data for searching and reporting (commonly based on OpenSearch).
- **Dashboard:** A web interface for visualizing alerts, managing agents, and reviewing compliance reports.

This distributed design enables scalability across thousands of monitored devices, from local systems to hybrid cloud environments. Wazuh architecture flow

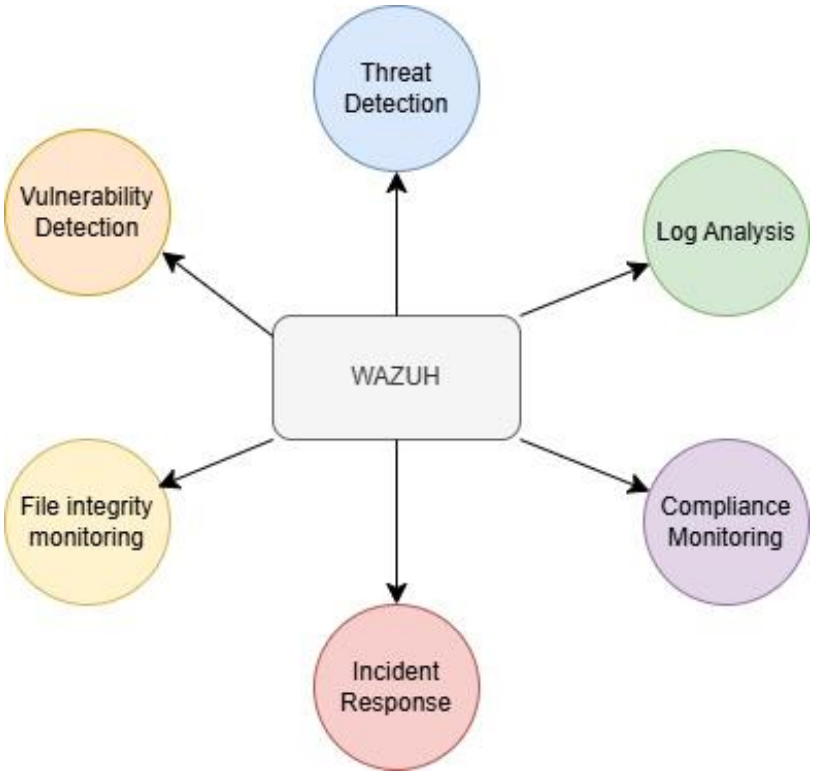


Core Capabilites

Wazuh provides features such as log data analysis, file integrity monitoring, rootkit detection, and active response to threats. It supports real-time event correlation, vulnerability assessment, and integration with third-party tools such as Elastic Stack and Splunk. Built-in modules facilitate compliance with standards like PCI DSS, HIPAA, and GDPR.

Uses and Deployment

Wazuh can be deployed on-premises, in hybrid setups, or within cloud environments like Amazon Web Services, Microsoft Azure, and Google Cloud Platform. It supports containerized installations using Docker or orchestration with Kubernetes, making it suitable for both small-scale and enterprise-level deployments.



Community and Development

Wazuh maintains an active open-source community with continuous development and updates. Its ecosystem includes documentation, integrations, and APIs, allowing customization and automation of security workflows. The platform is widely used by SOCs, managed security providers, and enterprises seeking a cost-effective SIEM and XDR solution.

Wazuh- Agent

The Wazuh Agent is a lightweight endpoint software component that collects security data from servers, workstations, and cloud instances for analysis by a central Wazuh manager. It plays a key role in monitoring and protecting diverse environments within the Wazuh Security Information and Event Management (SIEM) ecosystem.

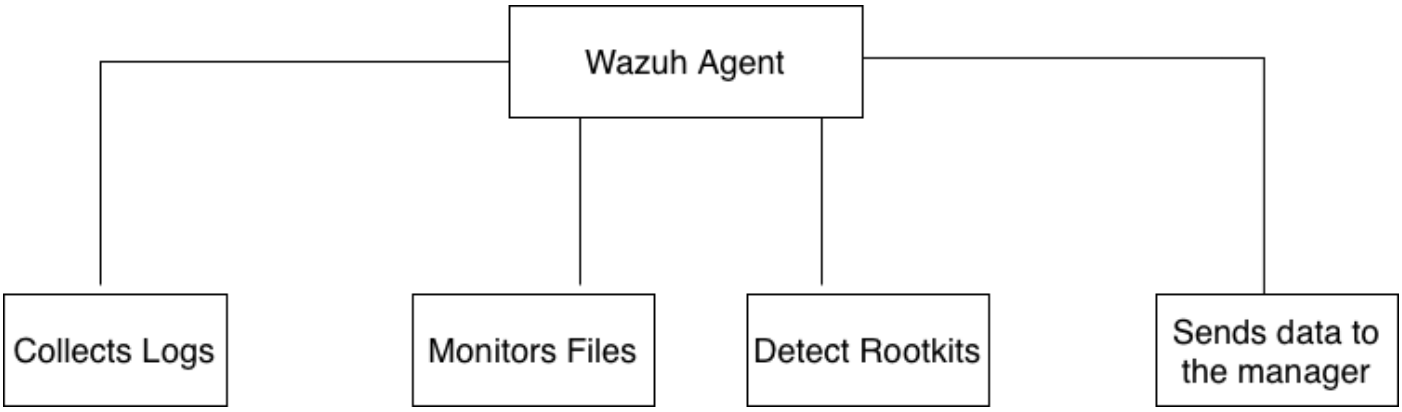
It communicates with the Wazuh server, sending data in near real-time through an encrypted and authenticated channel. The Wazuh agent provides capabilities such as log data collection, file integrity monitoring, threat detection, security configuration assessment, system inventory, vulnerability detection, and incident response to enhance your endpoint security.

Key facts

- Developer: Wazuh Inc.
- Initial release: 2015
- Primary function: Endpoint monitoring and data collection
- Supported platforms: Windows, Linux, macOS, Solaris, AIX, and BSD
- Integration: Communicates with Wazuh Manager via encrypted channels

Role and Functionality

The Wazuh Agent monitors the local system's activity, including logs, running processes, file integrity, rootkit detection, and active response actions. It gathers and normalizes data before sending it securely to the Wazuh Manager for centralized analysis. This helps organizations detect intrusions, misconfigurations, and compliance issues in real time.



Architecture and Communication

Agents operate as part of a distributed architecture, where each endpoint agent communicates with a Wazuh Manager or through intermediate relays. The communication uses a secure protocol (typically over TCP/UDP with optional TLS encryption). Agents can be remotely managed, allowing updates and configuration changes from the central management interface.

Deployment and Management

Agents can be deployed manually, via configuration management tools (like Ansible, Puppet, or Chef), or through the Wazuh web user interface. Each agent registers with the manager using unique authentication keys, ensuring trust and data integrity. Once registered, the agent's status and activity can be monitored through the Wazuh Dashboard or API.

Use cases and benefits

Wazuh Agents enable visibility across hybrid and multi-cloud infrastructures. Common use cases include compliance monitoring (PCI DSS, HIPAA, GDPR), intrusion detection, vulnerability assessment, and incident response automation. Their open-source nature and scalability make them a cost-effective choice for both enterprise and open environments.

Wazuh agent Lifecycle

The Wazuh agent lifecycle refers to the various stages a Wazuh agent goes through from its initial installation on an endpoint to its removal from the Wazuh platform. It includes the following stages:

1. Installation and enrollment
2. Agent connection states
3. Removal

Wazuh – Manager

The Wazuh manager is responsible for data analysis and alerting. It can forward alerts through syslog, emails, or integrated external APIs. See the data analysis documentation for more information on how Wazuh performs data analysis.

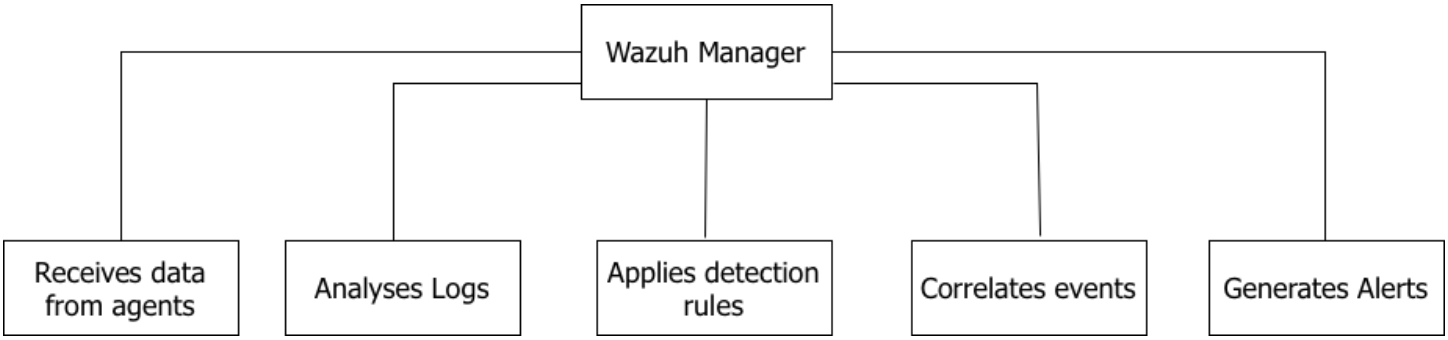
The Wazuh manager comprises several services and components that are responsible for various functions. These include enrolling new Wazuh agents, aggregating security events, decoding logs, evaluating rules, and alerting. It is also responsible for other functions like validating the Wazuh agent's identities and encrypting the communications between the Wazuh agent and the Wazuh server.

Key facts

- Primary role: Centralized event analysis and correlation server
- Core functions: Log analysis, rule evaluation, alert generation
- Deployment model: Agent-server with optional distributed clustering
- Integration: Works with Elastic Stack and OpenSearch
- License: Open source (GPL v2)

Architecture and role

The Wazuh Manager is responsible for receiving data from endpoint agents, decoding the data into structured formats, applying security rules, and generating alerts when anomalies or policy violations are detected. It maintains a rule set similar to OSSEC but extends it with advanced decoders and compliance checks. Multiple managers can form a cluster to provide scalability and high availability for large environments.



Core functions

Its main processes include log collection, rootcheck (for rootkit detection), file integrity monitoring (FIM), and active response. The manager continuously compares system states and events against policy rules to identify threats or configuration deviations. Alerts are stored locally and can be forwarded to indexing and visualization systems such as Kibana or OpenSearch Dashboards for real-time analysis.

Integration and management

Wazuh Manager integrates seamlessly with external SIEM or analytics tools through RESTful APIs and message queues. It can receive agent data over encrypted channels, manage agent registration, and provide configuration updates remotely. The Wazuh dashboard and RESTful API allow centralized visualization and operational management of alerts, inventory data, and security posture.

Use cases and importance

Enterprises and security teams use Wazuh Manager to centralize monitoring across servers, containers, cloud workloads, and endpoints. It supports compliance frameworks such as PCI DSS, HIPAA, and GDPR by automating log retention and audit reporting. Its flexibility and open architecture make it a cost-effective alternative to commercial SIEM products for unified threat detection and response.

Wazuh – Dashboard

Wazuh Dashboard is the web-based security visualization and management interface for the Wazuh XDR/SIEM platform. It provides a central place to visualize alerts, search security events, manage agents, and operate the overall Wazuh deployment through an interactive UI.

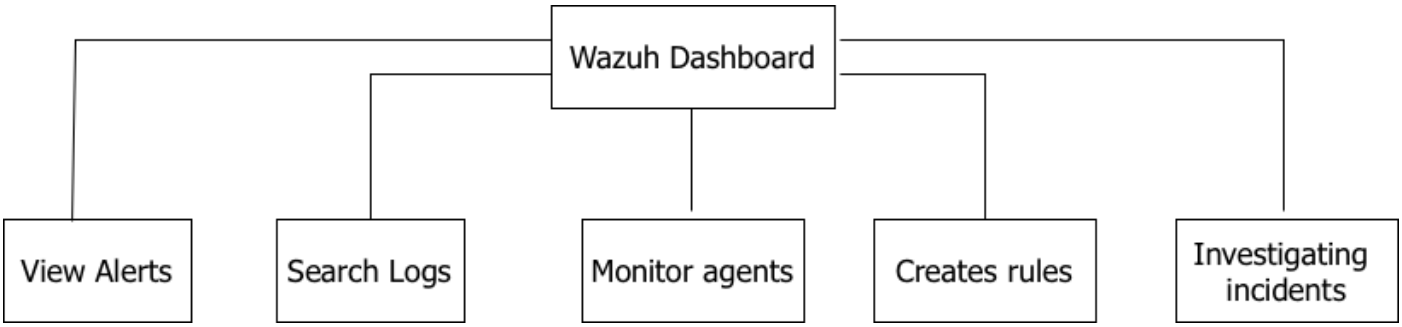
Key facts

- Type: Web UI / security visualization interface for Wazuh
- Built on: Fork of OpenSearch Dashboards
- Primary use: Threat detection, compliance monitoring, and security operations
- Scope: Visualizes data from Wazuh agents, servers, and indexer
- Licensing: Open-source, part of the Wazuh stack

Core purpose and capabilities

The dashboard is the main interface for analyzing alerts and security telemetry collected by Wazuh agents and agentless integrations. It offers prebuilt views for threat hunting, malware detection, file integrity monitoring (FIM), vulnerability status, and system inventory, as well as regulatory compliance (PCI DSS, GDPR, HIPAA, NIST 800-53, etc.).

Users can run ad-hoc searches, pivot across indices, and inspect raw events to investigate incidents. The visualization layer helps reveal trends and anomalies so analysts can move from high-level KPIs down to individual event details.



Data visualization and dashboards

Wazuh Dashboard includes many out-of-the-box dashboards plus the ability to build fully custom ones. Visualizations (charts, tables, maps, etc.) can be composed into dashboards for different use cases—SOC overview, authentication monitoring, endpoint health, compliance posture, and more.

Users can save searches, generate CSV reports, and use dedicated compliance modules (e.g., NIST 800-53, PCI DSS, HIPAA) that group alerts by control/requirement, making audits and continuous monitoring easier.

Operational and platform management

Beyond visualization, the dashboard is used to manage the Wazuh environment itself. From the UI, administrators can:

- View agent status and configure which modules/logs each endpoint collects
- Monitor the health, logs, and statistics of Wazuh components
- Create and test custom rules and decoders using built-in ruleset test tools
- Interact with the Wazuh server and indexer APIs via embedded consoles

This turns the dashboard into the primary control plane for security operations and platform administration in Wazuh-based environments.

User experience and ecosystem

The interface is browser-based and supports modern browsers like Chrome, Firefox, and Safari. It inherits much of the usability and plugin model from OpenSearch Dashboards, while adding Wazuh-specific apps, navigation, and branding. Organizations can also customize dashboards, branding, and SSL settings, and enable multi-tenancy to segregate data for different teams or customers.

Opensearch

OpenSearch is an open-source search and analytics engine designed for large-scale, real-time data exploration. Originating as a community-driven fork of Elasticsearch and Kibana, it provides a full-featured suite for indexing, querying, visualizing, and managing data. Maintained primarily by Amazon Web Services, OpenSearch is used for log analytics, application monitoring, and search-intensive applications.

Key facts

- Initial release: 2021
- Origin: Fork of Elasticsearch 7.10 and Kibana 7.10
- Primary language: Java
- License: Apache License 2.0
- Maintainer: OpenSearch community (led by AWS)

Background and development

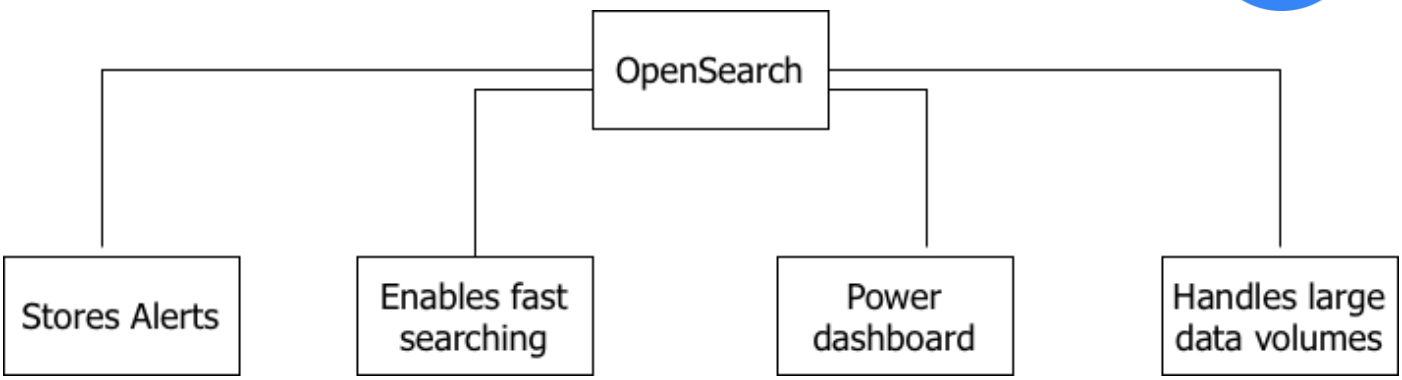
OpenSearch emerged in 2021 after Elastic changed the licensing of Elasticsearch and Kibana from Apache 2.0 to Server-Side Public License (SSPL). In response, AWS and the open-source community forked the last Apache-licensed versions to ensure an open, community-governed alternative. The project has since expanded beyond its origins, incorporating unique plugins, features, and governance structures to promote transparency and collaboration.

Core features

OpenSearch supports powerful full-text search, structured queries, and near real-time analytics. Its distributed architecture enables horizontal scaling across clusters. The accompanying visualization platform, OpenSearch Dashboards, allows interactive analysis of indexed data through charts and dashboards. The system includes plugins for machine learning, alerting, anomaly detection, and security management.

Use cases

Common applications include operational observability, log and metrics analysis, security information and event management (SIEM), and enterprise search. Organizations use OpenSearch to centralize diverse datasets, monitor application performance, and extract insights from large-scale logs and metrics in real time.



Governance and ecosystem

The OpenSearch Project operates under an open governance model with technical steering committees and community-driven development. It includes both the OpenSearch engine and OpenSearch Dashboards, with an ecosystem of plugins and integrations that extend its functionality across cloud and on-premises environments.